Adnan Mashrur Sadad

MLE Graduate — E-Commerce Risk Control (ByteDance)

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Summary

Built a fraud detection system that treats accuracy as a trap — 0.98 ROC-AUC and 0.89 PR-AUC on a 0.17% base rate, by pairing supervised ensembles with unsupervised anomaly detection and shipping both behind a real-time API. Looking for a risk problem with real transaction volume behind it, where "caught it before it cleared" is the only metric that counts.

Technical Skills

- **Risk & ML Systems:** Fraud & Anomaly Detection, Ensemble Classification (Random Forest, XGBoost, Isolation Forest), Class-Imbalance Handling (SMOTE, Class-Weighted Loss), PR-AUC / ROC-AUC Evaluation, Scikit-learn, Hugging Face, Real-Time Inference Serving (FastAPI, Pydantic)
- **GenAI / Agentic Systems:** RAG, Text2SQL, Structured Output Validation & Repair Loops, LLM Evaluation Harnesses, Agent Pipelines, Prompt Engineering, Groq API
- **Languages:** Python, Java, C++, JavaScript, TypeScript, SQL
- **Frameworks & Deployment:** FastAPI, SQLAlchemy, Streamlit, Docker, REST APIs, CI/CD (GitHub Actions), OpenAPI

Projects

AI Financial Fraud Detection System — Real-Time Risk Classifier 2026

Python, XGBoost, Scikit-learn, FastAPI, Streamlit

- **Caught fraud at a 0.17% base rate, where a model that always predicts "not fraud" scores 99.8% accuracy and 0% recall,** by rejecting accuracy as the headline metric and optimizing for PR-AUC (0.89) instead.
- **Reached 0.98 ROC-AUC / 0.89 PR-AUC on held-out data,** by ensembling Random Forest, XGBoost, and unsupervised Isolation Forest so the system flags both known fraud patterns and behaviour that matches no labeled case at all.
- **Corrected for extreme class imbalance without discarding minority-class signal,** by combining SMOTE oversampling with class-weighted loss across 284,807 real credit-card transactions.
- **Turned the model into something an analyst could act on,** by serving it as a FastAPI inference endpoint with Pydantic validation and a Streamlit dashboard for batch scanning and explainability.

InterviewPilot — Production LLM Agent Pipeline 2026 – GitHub

Python, FastAPI, SQLAlchemy, React/TypeScript, Groq API, Docker, GitHub Actions

- **Closed the gap between raw LLM output and a machine-readable contract,** by building a self-correcting repair loop that catches schema-invalid model output and re-prompts until it validates.
- **Made agent behavior safe to change without live inference cost,** by building a 17-test regression harness at 87% backend coverage with model calls fully mocked, gated in CI on every push.

Mindhive Chatbot — Multi-Turn Agentic Pipeline 2025 – GitHub

FastAPI, RAG, Text2SQL

- **Held coherent context across 3–5 conversational turns,** by designing stateful, intent-based planning instead of stateless per-message handling.
- **Proved retrieval quality before shipping,** with an 85% Hit Rate@3 (100% on outlet lookups) and 0.86 MRR on a hand-labeled 20-query evaluation set.

Work Experience

Joget Inc. Oct 2024 – Mar 2025

Software Developer Intern (Hybrid) *Kuala Lumpur, MY*

- **Cut manual processing effort by 30% on a core client workflow,** by integrating REST APIs between Joget DX and external services to automate a previously manual data flow.
- **Shipped production features inside real sprint cycles,** by building and deploying enterprise workflow automations and UI components on the Joget DX low-code platform, from design through CI/CD-managed release.

Education

Universiti Teknologi Malaysia (UTM), MJIIT 2021 – 2026

Bachelor of Software Engineering, CGPA: 3.50/4.00 — Dean's List 2022/2023 — Top Performer, Computational Intelligence (2023)*

Kuala Lumpur, Malaysia

*Final CGPA pending official publication.

Achievements

- Gold Medal & Best Video Award, myHCI-UX Student Design Challenge (Oct 2025) — national university competition for UX design and innovation
- Secretary Treasurer, SOFEA Society (2022–2023); organized AI workshops for 200+ students